

JEE ADVANCED WEEKEND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA



Sec: SR.IIT_*CO-SC(MODEL-A&B)
Time: 3 Hrs

Date: 14-07-24
Max. Marks: 180

14-07-24_SR.IIT_STAR CO-SC(MODEL-A&B)_JEE ADV_WAT-54_SYLLABUS

MATHEMATICS: **PRESENT WEEK(80%): Probability:** Classical definition of probability, types of events, Addition & Multiplication theorem, Problems based on P & C

PREVIOUS WEEK(20%): 100% COMPLETE P&C

PHYSICS: **PRESENT WEEK(80%): MODEREN PHYSICS: Matter waves:** Debroglie's wavelength, Photo Electric Effect (**Exclude: Davisson Germer experiment**)

PREVIOUS WEEK(20%): Diffraction due to a single slit, Polarization of light, plane polarized light; Brewster's law, Polaroids.

(EXCLDE: Resolving Power)

EM Waves: Electromagnetic waves and their characteristics, Transverse nature of electromagnetic waves, Electromagnetic spectrum (radio waves, microwaves, infrared, visible, ultraviolet, X-rays, Gamma rays), Applications of e.m. waves.

CHEMISTRY: **PRESENT WEEK(80%): Qualitative Analysis:** Introduction, flame test, borax bead test, charcoal test, Acid radicals- carbonate, bicarbonate, sulphite, thiosulphate, nitrite, nitrate, chloride, bromide, iodide, sulphate and phosphite only

PREVIOUS WEEK(20%): Bonding in metal carbonyls, structures of mononuclear & dinuclear metal carbonyls (NCERT), CFT (octahedral and tetrahedral fields)], spectro chemical series, Stability, importance and applications of complexes, isomerism in complexes

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 6)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II(Q.N : 7 – 14)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+3	0	8	24
Sec – III(Q.N : 15-18)	Questions with Comprehension Type (2 Comprehensions – 2 +2 = 4Q)	+3	-1	4	12
Total				18	60

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N :19 – 24)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II(Q.N : 25 –32)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+3	0	8	24
Sec – III(Q.N : 33-36)	Questions with Comprehension Type (2 Comprehensions – 2 +2 = 4Q)	+3	-1	4	12
Total				18	60

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N:37 – 42)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II(Q.N :43–50)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+3	0	8	24
Sec –III(Q.N : 51-54)	Questions with Comprehension Type (2 Comprehensions – 2 +2 = 4Q)	+3	-1	4	12
Total				18	60

THE NARAYANA GROUP

SECTION- I

(Maximum Marks : 24)

This section contains SIX (06) questions.

Each question has FOUR options for correct answer(s). ONE OR MORE THAN ONE of these four option(s) is (are) correct option(s).

For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks: -2 In all other cases.

1. For persons A,B,C,D roll a die each and the numbers obtained by them are noted as x_1, x_2, x_3, x_4 respectively . The probability that

A) $x_1 < x_2 < x_3 < x_4$ is $\frac{5}{432}$

B) $x_i \leq x_j$ for $i < j$ is $\frac{7}{72}$

C) $x_1 + x_2 + x_3 + x_4 = 10$ is $\frac{80}{1296}$

D) $x_1 < x_2$ and $x_3 < x_4$ is $\frac{25}{144}$

2. A box contains 15 cards numbered 1 to 15. Five cards drawn at random one after the other with replacement the numbers obtained from the cards drawn are noted. The probability that

A) The largest number is at most 7 is $\left(\frac{7}{15}\right)^5$

B) The largest number is 7 is $\frac{7^5 - 6^5}{15^5}$

B) The smallest number is at least 9 is $\left(\frac{7}{15}\right)^5$

D) The smallest number is 9 is $\frac{16}{1215}$

3. A set A contains 15 elements. A subset P of A is selected at random. After replacing the elements, again a subset Q of A is selected. Which of the following are correct?
- A) The probability that $P \cap Q = \phi$ is $\left(\frac{3}{4}\right)^{15}$
- B) The probability that $P \cup Q = A$ is $\left(\frac{3}{4}\right)^{15}$
- C) The probability that $P \cap Q$ has exactly 3 elements is $\frac{15C_3 \times 3^{12}}{4^{15}}$
- D) The probability that $P \cup Q$ has exactly 3 elements is $\frac{15C_3 \times 3^{12}}{4^{15}}$
4. 3 whole numbers are randomly chosen one by one with replacement and multiplied. Then which of the following is true
- A) The probability that the last digit is 1, 3, 7 or 9 is $\frac{4^3}{10^3}$
- B) The probability that the last digit is 2, 4, 6, 8 is $\frac{8^3 - 4^3}{10^3}$
- C) The probability that the last digit is 5 is $\frac{5^3 - 4^3}{10^3}$
- D) The probability that the last digit is zero is $\frac{10^3 - 8^3 + 5^3 + 4^3}{10^3}$
5. Probability of n heads in 2n tosses of a fair coin can't be given by
- A) $\prod_{r=1}^n \left(\frac{2r-1}{2r}\right)$ B) $\prod_{r=1}^n \left(\frac{n+r}{2r}\right)$ C) $\sum_{r=0}^n \left(\frac{{}^nC_r}{2^n}\right)$ D) $\sum_{r=0}^{2n} {}^{2n}C_r$
6. A bag contains b blue and r red balls. If two balls are drawn at random, the probability of drawing two red balls is five times the probability of drawing two blue balls. Furthermore, the probability of drawing one ball of each colour is six times the probability of drawing two blue balls. Then
- A) $b + r = 9$ B) $br = 18$ C) $|b - r| = 4$ D) $b / r = 2$

SECTION - II**(Maximum Marks : 24)**

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

7. Functions are formed from $A = \{1, 2, 3\}$ to set $B = \{1, 2, 3, 4, 5\}$ and one function is selected at random. If P the probability that function satisfying $f(i) \leq f(j)$ whenever $i < j$ then value of $25P$ is equal to
8. A bag contains n identical red balls, $2n$ identical black balls and $3n$ identical white balls. N balls are drawn at random and kept on a table. The colour combinations of the balls are observed. If the probability that the combination includes at least one ball of each colour is $\frac{1}{5}$ then $n =$
9. If the sides of triangle are decided by throwing a die thrice, the probability that the triangle is isosceles or equilateral is $\frac{1}{k}$ then $k =$
10. An unbiased coin is tossed 12 times. The probability that at least 7 consecutive heads show up is $\frac{K}{256}$ then $K =$
11. Let $s = \{1, 2, 3, \dots, n\}$, If X denote the set of all subsets of S containing exactly two elements, then the value of $\sum_{A \in X} (\min A)$ is ${}^{n+1}C_\lambda$ then $\lambda =$ _____
12. A die is weighted such that the probability of rolling the face numbered n is proportional to n^2 ($n = 1, 2, 3, 4, 5, 6$). The die is rolled twice, yielding the numbers a and b . The probability that $a < b$ is p , then the value of $[2/p]$ is (where $[.]$ represents the greatest integer function)
13. Consider the system of equations $ax + by = 0$, $cx + dy = 0$, where $a, b, c, d \in \{0, 1\}$. The probability that the system of equations has a unique solution is k then $8k$ is _____
14. Two different numbers are taken from the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. The probability that their sum and positive difference are both multiple of 4 is $x/55$, then x equals _____

SECTION – III (COMPREHENSIN TYPE)

This section contains 2 Paragraphs. Based on each paragraph, there are 2 questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Paragraph for Question Nos. 15 to 16:

If A, B are two events then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

15. Consider the statements

i) For events $A_1, A_2, \dots, A_n$,

$$\max\left(0, 1 - n + \sum_{k=1}^n P(A_k)\right) \leq P(A_1 \cap A_2 \cap \dots \cap A_n) \leq \min(P(A_1), P(A_2), \dots, P(A_n))$$

ii) For the events $A_1, A_2, \dots, A_n$,

$$\max(P(A_1), P(A_2), \dots, P(A_n)) \leq P(A_1 \cup A_2 \cup \dots \cup A_n) \leq \min\left\{1, \sum_{k=1}^n P(A_k)\right\}$$

A) Both statements (i), (ii) are true

B) Statement (i) is true (ii) is False

C) Statement (i) True (ii) is false

D) Statement (i) is false (ii) is false

16. In a battel 70% of the combatants lost one eye, 80% an ear, 75% an arm, 85% a leg, x % lost all four limbs minimum value of x is

A) 10

B) 12

C) 15

D) None

Paragraph for Question Nos. 17 to 18:

Let A be an event then probability happing event A is equal to the ratio of number of favorable cases to A to total number of cases

17. Two squares are chosen at random from the smallest squares on a chess board. What is the chance that the two squares chosen have exactly one corner in common?

A) $\frac{7}{144}$

B) $\frac{12}{144}$

C) $\frac{5}{144}$

D) None

18. A man can take a step forward, backward, left or right with equal chance. What is the probability that after 9 steps he will be just one step away from his initial position?

A) $\frac{3969}{4^7}$

B) $\frac{3967}{4^8}$

C) $\frac{3965}{4^9}$

D) None

SECTION- I

(Maximum Marks : 24)

This section contains SIX (06) questions.

Each question has FOUR options for correct answer(s). ONE OR MORE THAN ONE of these four option(s) is (are) correct option(s).

For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

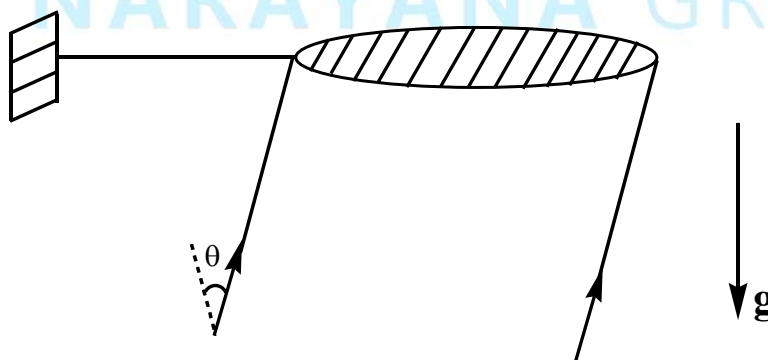
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks: -2 In all other cases.

19. Consider white light whose wavelength spread is from 400nm to 700nm. Its energy is uniformly distributed in this spectrum. The light is incident on a metal A of work function 1.55eV. Saturation photo current is 6mA. Now the same light is incident on metal B of work function 2.48eV. Choose the correct options.

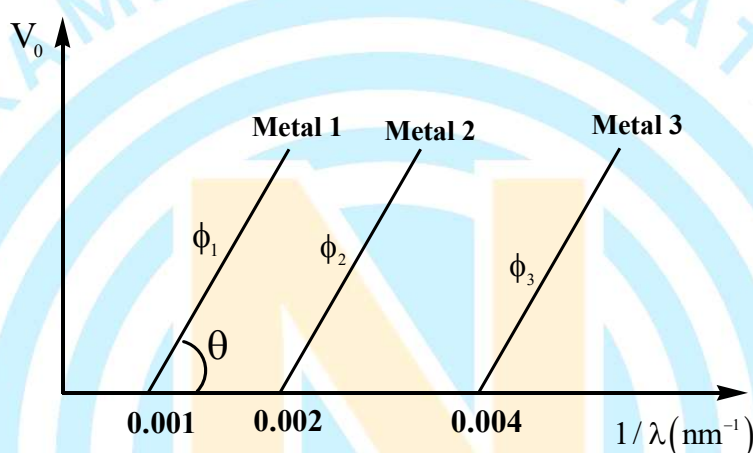
[Take $hc = 1240 \text{ eV nm}$. Assume photo efficiency remains same]

- A) Stopping potential for experiment with metal A is 0.22 V
 B) Stopping potential for experiment with metal B is 0.62 V
 C) Saturation photo current for metal B will be 2mA
 D) Saturation photo current for metal B will be $\frac{18}{11} \text{ mA}$
20. Consider the arrangement shown in figure. Disc is uniform of mass m and of radius R . Intensity of parallel beam of falling on disc is uniform. Disc is in equilibrium due to radiation falling on it. String is ideal and coefficient of absorption for disc is $\frac{1}{3}$ and it does not transmit any light. Choose the correct option(s): (Plane of disc is horizontal)



- A) Tension in the string is $T = \frac{mg \tan \theta}{5}$
- B) Tension in the string is $T = \frac{2mg \tan \theta}{5}$
- C) Intensity of the beam is $I = \frac{2mgC}{5\pi R^2 \cos^2 \theta}$
- D) Intensity of the beam is $I = \frac{3mgC}{5\pi R^2 \cos^2 \theta}$

21. The graph between the stopping potential (V_0) and $(1/\lambda)$ is shown in Fig. ϕ_1 , ϕ_2 and ϕ_3 are work functions. Which of the following is/are correct?



- A) $\phi_1 : \phi_2 : \phi_3 = 1 : 2 : 4$
- B) $\phi_1 : \phi_2 : \phi_3 = 4 : 2 : 1$
- C) $\tan \theta$ is directly proportional to hc/e , where h is Planck's constant and c is the speed of light and e is charge of electron.
- D) Ultraviolet light can be used to emit photoelectrons from metal 2 and metal 3 only
22. Photoelectric effect supports quantum nature of light because:
- A) there is a minimum frequency of light below which no photoelectrons are emitted
- B) the maximum kinetic energy of photoelectrons depends only on the frequency of light and not on its intensity
- C) even when the metal surface is faintly illuminated, the photoelectrons leave the surface immediately
- D) electric charge of the photoelectrons is quantized

23. In which of the following situations, the heavier particles has smaller de Broglie wavelength compared to light particles? (When all particles of different masses)
- A) move with the same speed
 - B) move with the same linear momentum
 - C) move with the same kinetic energy
 - D) have fallen through the same height
24. A parallel beam of fast moving electrons is incident normally on a narrow slit. A fluorescent screen is placed at a large distance from the slit. If the speed of the electrons is increased, which of the following statements is correct? The wave length of electron wave is equal to its De Broglie wavelength
- A) Diffraction pattern is not observed on the screen in the case of electrons.
 - B) The angular width of the central maximum of the diffraction pattern will increase
 - C) The angular width of the central maximum will decrease.
 - D) The angular width of the central maximum will be unaffected.

SECTION - II**(Maximum Marks : 24)**

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.

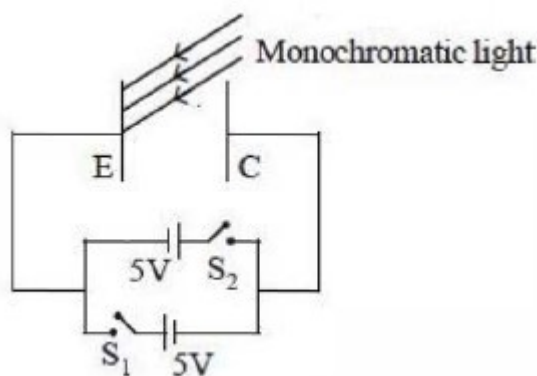
Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If **ONLY** the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

25. A point source of light emits photons of energy 4.8 eV at the rate 10^5 photons per second. These photons incident on a photo-sensitive sphere of work function 2.8 eV and radius 9 mm. The sphere is initially neutral and the emitted photo electrons are instantly swept away. The time after which the photo electric emission stops is (in seconds)
- (Assume 100% photo efficiency for electron emission)
26. Photons of wavelength λ emitted by a source of power P incident on a photo cell. If the current produced in the cell is I. the percentage of incident photons which produce current in the photo cell is $\frac{5x(Ihc)}{eP\lambda}$, where 'x' is ____ (h is Planck's constant and c is the speed of light in vacuum)
- (Assume 100% photo efficiency for electron emission)

27. A photodiode sensor is used to measure the output of a 300W lamp kept 10m away. The sensor has an opening of 2 cm in diameter. How many photons in multiples of 10^{10} enter the sensor if the wavelength is 660nm and the exposure time is 100ms (Assume that all the energy of the lamp is given off as light and $h = 6.6 \times 10^{-34} \text{ J-s}$)
28. The maximum kinetic energy of a photo electron(in eV) liberated from the surface of lithium with work function 2.35eV by electromagnetic radiation whose electric component varies with time as $E = a \left[1 + \cos(2\pi f_1 t) \right] \cos 2\pi f_2 t$ (where 'a' is a constant) is
($f_1 = 3.6 \times 10^{15} \text{ Hz}$, $f_2 = 1.2 \times 10^{15} \text{ Hz}$ and Plank's constant $h = 6.6 \times 10^{-34} \text{ J-s}$)
29. In a photoelectric experiment, a monochromatic light is incident on the emitter plate E, as shown in the figure. When switch S_1 is closed and switch S_2 is open, the photoelectrons strike the collector plate C with a maximum kinetic energy of 1 eV. If switch S_1 , is open and switch S_2 is closed and the frequency of the incident light is doubled, the photoelectrons strike the collector plate with a maximum kinetic energy of 20 eV. The threshold wavelength of the emitter plate is in $\text{\AA}$ (Take $hc=1242\text{eVnm}$)



30. Assume that the de-Broglie wave associated with an electron can form a standing wave between the atoms arranged in a one-dimensional array with nodes at each of the atomic sites. It is found that one such standing wave is formed if the distance d between the atoms of the array is $1.2 \text{\AA}$. A similar standing wave (of same wave length) is again formed if d is increased to $1.5 \text{\AA}$ but not for any intermediate value of d . Least value of d in $\text{\AA}$ is equal to

31. An electron of charge 'e' and mass 'm' moving with an initial velocity $v_0 \hat{i}$ is subjected to an electric field $E_0 \hat{j}$. The de-Broglie wavelength of the electron at a time 't' is (Initial de-Broglie wave length of the electron = λ_0) $\lambda_0 \left[1 + \frac{(eE_0 t)^x}{(mv_0)^y} \right]^z$, then $x + y + z$ is ____
32. The energy of a photon is equal to the kinetic energy of a proton. If λ_1 is the de Broglie wavelength of a proton, λ_2 is the wavelength associated with the photon, and if the energy of the photon is E, then (λ_1 / λ_2) is proportional to E^x . Where x is ____

SECTION – III
(COMPREHENSIN TYPE)

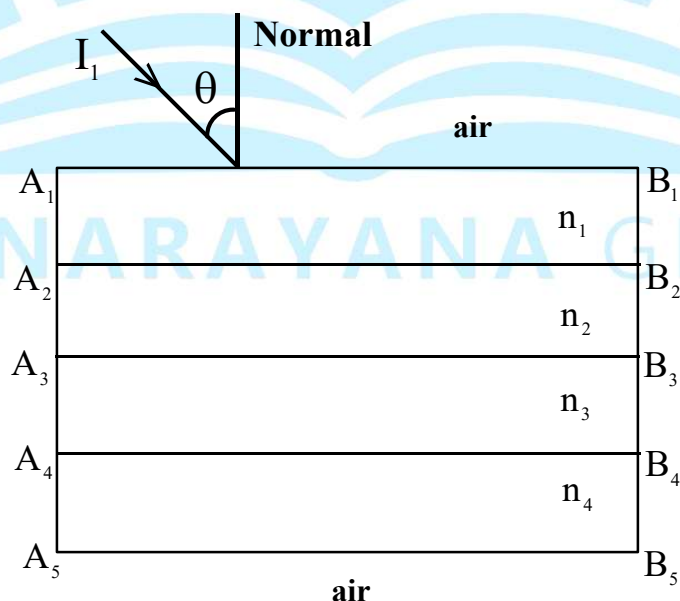
This section contains 2 Paragraphs. Based on each paragraph, there are 2 questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Paragraph for Question Nos. 33 to 34:

Figure shows 4 transparent parallel slabs of refractive indices n_1, n_2, n_3 & n_4 and I_1 is the incident unpolarised light. Given that $n_1 = \frac{6}{5}$, $n_2 = \frac{4}{3}$, $n_3 = \frac{3}{2}$ and $n_4 = \frac{8}{5}$

θ is the angle of incidence of incident ray falling from air onto a slab of refractive index n_1 .



33. The value of θ for which reflected ray becomes completely plane polarized at interface $A_2 B_2$ is

A) $\theta = \sin^{-1}\left(\frac{12}{\sqrt{181}}\right)$

B) $\theta = \sin^{-1}\left(\frac{12}{\sqrt{145}}\right)$

C) θ does not exist

D) $\theta = \sin^{-1}\left(\frac{6}{\sqrt{61}}\right)$

34. The value of θ for which reflected ray becomes completely plane polarized at interface $A_3 B_3$ is

A) $\theta = \sin^{-1}\left(\frac{12}{\sqrt{181}}\right)$

B) $\theta = \sin^{-1}\left(\frac{12}{\sqrt{145}}\right)$

C) θ does not exist

D) $\theta = \sin^{-1}\left(\frac{6}{\sqrt{61}}\right)$

Paragraph for Question Nos. 35 to 36:

In performing photoelectric experiment to study photoelectric effect, intensity of radiation (I), frequency of radiation (ν), work function (ϕ_0) of the photosensitive emitter, distance (d) between emitter and collector are changed or kept constant.

(I meant for number of photons incident per unit time)

35. If ν is increased, keeping I , ϕ_0 , d constant

A) Saturation photoelectric current increases

B) Stopping potential (V_0) decreases

C) Maximum KE ($K_{\max}$) of photoelectrons increases

D) Stopping potential remains the same

36. If d is increased, keeping I , ν , ϕ_0 constant

A) Saturation photoelectric current increases

B) Stopping potential (V_0) increases

C) Maximum KE ($K_{\max}$) of photoelectrons increases

D) Stopping potential remains the same

SECTION- I

(Maximum Marks : 24)

This section contains SIX (06) questions.

Each question has FOUR options for correct answer(s). ONE OR MORE THAN ONE of these four option(s) is (are) correct option(s).

For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks: -2 In all other cases.

37. Which of the following react(s) with $\text{dil. H}_2\text{SO}_4$?

- A) CaCO_3 B) KNO_2 C) Na_2S D) ZnSO_3

38. Borax bead test is given by

- A) Co^{3+} B) Zn^{2+} C) Cu^{2+} D) Ni^{2+}

39. Which of the following will be completely or partially dissolved in NH_4OH ?

- A) AgCl B) AgBr C) Ag_2CO_3 D) AgI

40. Which of the following statement(s) is(are) correct when a mixture of NaCl and $\text{K}_2\text{Cr}_2\text{O}_7$ is gently warmed with cone. H_2SO_4 ?

- A) Red vapour is evolved
 B) The vapours when passed into NaOH solution gives a yellow solution of Na_2CrO_4
 C) Chlorine gas is evolved
 D) Chromyl chloride is formed

41. which of the following statements is/are TRUE?

A) Like CO_3^{2-} , SO_3^{2-} also gives test with baryta water, $\text{Ba}(\text{OH})_2$.

B) $\text{Na}_2\text{S}_{(\text{aq})} + \text{Sodium nitro prussite} \rightarrow \text{Complex}$

In this reaction, change in oxidation state of iron from reactant to product complex is zero

C) Brown ring test is given by NO_3^- and not by SO_3^{2-}

D) Ag_2SO_3 is insoluble in dilute HNO_3 .

42. In which of the following pair(s) both the complexes show optical isomerism ?

- A) $cis-[Cr(C_2O_4)_2Cl_2]^{3-}$, $mer-[Co(gly)_3]$
- B) $[Co(en)_3]Cl_3$, $cis-[Co(en)_2Cl_2]Cl$
- C) $[PtCl(dien)]Cl$, $[NiCl_2(PPh_3)_2]$
- D) $[Co(NO_3)_3(NH_3)_3]$, $cis-[Pt(en)_2Cl_2]Cl_2$

SECTION - II

(Maximum Marks : 24)

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

43. A metal complex having composition $Cr(NH_3)_4Cl_2Br$ has been isolated in two forms (A) and (B). The form A reacts with $AgNO_3$ to give a white ppt, readily soluble in dilute aqueous ammonia, whereas (B) gives a pale yellow precipitate partial soluble in concentrated ammonia. Calculate magnetic moment (in B.M.) of B (spin – only value) (round off to nearest integer)

44. Calculate the difference in net stabilization energies (in kJ mol^{-1}) of Fe^{2+} complexes with CN^- ligands ($\Delta_0 = 25000 \text{ cm}^{-1}$) and with H_2O ligands ($\Delta_0 = 10000 \text{ cm}^{-1}$). Given that the pairing energy, $P = 15000 \text{ cm}^{-1}$ for Fe^{2+} ion for both the complexes.

(Given that $h = 6 \times 10^{-34} \text{ J sec}$., Avogadro Number = 6×10^{23} , Speed of light in vacuum = $3 \times 10^8 \text{ m sec}^{-1}$).

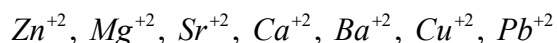
45. $Fe^{2+}(aq) + NO_3^-(aq) + H_2SO_4(\text{conc.}) \longrightarrow$ Brown ring complex

What is the atomicity of the entire complex?

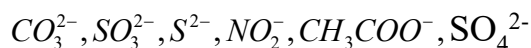
46. Number of the following metal cation(s) produce coloured bead in microcosmic salt bead test

Al^{3+} , Pb^{+2} , Zn^{+2} , Cu^{+2} , Cr^{3+} , Mn^{2+} , Ba^{2+} , Fe^{3+}

47. How many of the following nitrate salts give paramagnetic gas(es) on heating



48. Number of anions which evolve gases on reaction with *dil. HCl* / *dil. H₂SO₄*.



49. The number of option(s) regarding the complex $[Co(en)(NH_3)_3(H_2O)]^{3+}$

(*en* = $H_2NCH_2CH_2NH_2$) are

A) It has two geometrical isomers

B) It will have three geometrical isomers if bidentate 'en' is replaced by two cyanide ligands

C) It is paramagnetic

D) It absorbs light at longer wavelength as compared to $[Co(en)(NH_3)_4]^{3+}$

50. Out of total P anions, if Q anions give colourless acid vapour/gas with *conc. H₂SO₄* on reaction with following given anion.



Find value of $\frac{P}{Q}$

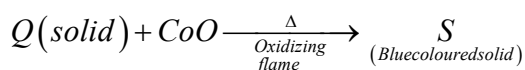
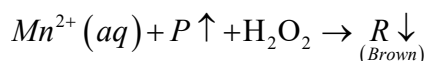
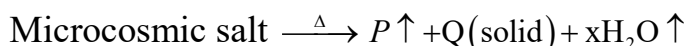
SECTION – III (COMPREHENSIN TYPE)

This section contains 2 Paragraphs. Based on each paragraph, there are 2 questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Paragraph for Question Nos. 51 to 52:

For following reactions:



51. **Incorrect** statement for compound 'S' is:

- A) It is paramagnetic.
- B) It is orthophosphate.
- C) Its colour remains same even in reducing flame
- D) It is reduced to metal cobalt in reducing flame.

52. Which of the following reaction(s) is relevant to microcosmic salt bead test?

- A) $Cr_2O_3 + 3B_2O_3 \rightarrow 2Cr(BO_2)_3$
- B) $CoO + ZnO \rightarrow CoZnO_2$
- C) $CoO + NaPO_3 \rightarrow NaCoPO_4$
- D) $Al_2(SO_4)_3 + 3Na_2CO_3 \rightarrow Al_2O_3 + 3Na_2SO_4 + 3CO_2$

Paragraph for Question Nos. 53 to 54:

Solid 'X' salt + MnO_2 (Solid)

↓
conc. H_2SO_4 added dropwise
and on warming

↓
'Y' (coloured gases)

↓
+ [KI – starch paper]

↓
[Turns KI – starch paper blue]

53. Here salt 'X' maynot be:

- A) $NaCl$
- B) $NaBr$
- C) NaI
- D) None of these

54. Which of the following statements is **CORRECT**?

- A) MnO_2 oxidizes halides ions into dihalogen
- B) Reducing nature of halides ions follow the order, $F^- < Cl^- < Br^- < I^-$
- C) 'Y' gas can be I_2 , which turns KI-starch paper into blue colour.
- D) 'Y' gas can be Cl_2 and Br_2 , which oxidises of KI to I_2 & turns starch paper blue.